SORT appliance by Priority

For each appliance DO

IF electricity_tariff = Low
then
Schedule appliance during off-peak hours

ELSE
If total_load + power consumption ≤ max_load
then
Schedule appliance
update total_load

Else
postpone appliance

Display appliance Schedule
Calculate Electricity_Cost
Display Electricity_Cost

o/p seen Sruthy 09/10/26

output :

- - - Appliance scheduling order - - -
00:00 - 01:00 - Refrigirator, AC, Washing mac
01:00 - 02:00 - Refrigirator, AC, Washing machine
02:00 - 03:00 - Refrigirator, AC, Washing machine
03:00 - 04:00 - Refrigirator, AC, Washing machine
04:00 - 05:00 - Refrigirator, AC
05:00 - 06:00 - Refrigirator AC
06:00 - 07:00 - Refrigirator
07:00 - 08:00 - Refrigirator
08:00 - 09:00 - Refrigirator

Smart Electricity Load Scheduling

Statement:
Smart Electricity Load Scheduling is a system that efficiently manages the electricity consumption of household appliances. It schedules appliances based on their priority, electricity tariff and available power to reduce electricity bills and avoid peak-hour loads

Objective :-
To develop a smart electricity load scheduling system that optimizes electricity consumption by scheduling appliances based on priority, tariff and power limits, thereby reducing electricity costs and peak-hour load.

Pseudocode :

// Input number of Appliances.

For each appliances
    // Input name, power_consumption, priority, operating time

// Input electricity tariff, maximum load

Date __/__/__
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Genetic optimization

Algorithm:

Step1: Generate a population of chromosomes randomly

Step2: Fitness Evaluate: Using fitness function evaluate each chromosome

Step3: Select the parents with higher fitness value

Step4: ~~Select two or more pair~~ Do crossover for the parents to produce offsprings

Step-5: Apply random changes to offspring genes to maintain diversity i.e. mutation
Replace old Population with new population

Step-6: Move to Step 3 if criteria is satisfying fitness offspring found. otherwise terminate

Step7: Return the best offspring obtained during evolution